

4.PROJECT PLANNING PHASE

The Project Planning Phase determines how the project will be implemented and organized. The entire development process is divided into smaller tasks so that each component can be developed and tested systematically.

4.1 Development Plan

The development plan consists of the following activities:

Step 1 – Environment Setup

Python and Visual Studio Code are installed and configured. A Python virtual environment is created to isolate project dependencies.

Step 2 – Backend Development

FastAPI is configured and the basic application structure is created.

Step 3 – Database Development

SQLAlchemy models are created for storing users and workout plans.

Step 4 – Frontend Development

Jinja2 templates are developed for collecting information and displaying AI-generated results.

Step 5 – Gemini Integration

The Gemini API is integrated into the application to generate personalized workout plans and nutrition guidance.

Step 6 – Feedback Implementation

A feedback mechanism is implemented to allow users to request modifications to their workout plans.

Step 7 – Administrator Module

A protected administrator interface is created to display users and their plans.

Step 8 – API Development

REST API endpoints are created for workout generation and feedback processing.

Step 9 – Testing

The complete system is tested at unit, integration, API, and user-interface levels.

Step 10 – Documentation

The implementation process, architecture, requirements, testing results, and usage instructions are documented.

4.2 Project Schedule

A simple development schedule can be organized as:

Phase	Main Activity
Week 1	Brainstorming and requirement analysis
Week 2	System design and planning
Week 3	Backend and database development
Week 4	Frontend development
Week 5	Gemini AI integration
Week 6	Feedback and admin module
Week 7	Testing and debugging
Week 8	Documentation and demonstration

This planning structure helps ensure that the project is developed systematically.